Thematic Analysis Tool for Health Providers Technical Specification

1. Introduction

1.1 Overview

Patient and public survey data are crucial for shaping health services, but analysing qualitative (free-text) feedback requires significant manual effort. While existing AI tools can generate themes rapidly, they often act as "black boxes," failing to provide the transparency needed to verify how conclusions were drawn. This lack of transparency leads to unchecked bias, reduced trust, and difficulty tracing themes back to patient voices.

This project aims to develop a transparent, interactive analysis tool that transforms overwhelming volumes of qualitative patient feedback into actionable insights. The tool features a visual, drag-and-drop workspace that mimics physical post-it note clustering. Users initially organise central quotes into manual clusters, teaching the AI the specific patterns and nuances of the data. Once enough quotes are clustered manually, the AI assists by suggesting theme names and automatically assigning new, incoming survey responses to the established clusters.

1.2 Objectives

The system is designed to:

- Process large volumes of unstructured qualitative survey feedback.
- Provide an intuitive, visual drag-and-drop interface for manual quote clustering.
- Utilise AI to suggest theme names based on user-created clusters.
- Automatically categorise incoming future survey responses based on learned patterns.
- Provide a transparent audit trail linking every theme back to specific, anonymised patient quotes.

1.3 Scope (MVP)

- Visual Drag-and-Drop Clustering Interface (Post-it Note UI).
- AI Theme Name Generation.
- AI Auto-Assignment for new quotes.
- Quote Traceability (Evidence Linking).
- Metadata Integration.
- Secure, In-House Processing.

2. System Architecture

2.1 High-Level Architecture

The system follows a modular architecture built upon the *richardgomer/nie-prototype.*

Frontend (React - Drag-and-Drop Visual Workspace)
↓
FastAPI Backend (Python - Survey Data & State Management)
↓
NLP Pipeline (Python - Theme Naming & Auto-Classification)
↓
Secure In-House Database (Strict NHS Compliance)

3. System Components

3.1 Frontend (React)

The frontend serves as an interactive Human-in-the-Loop (HITL) playground.

Features:

- Central Quote Repository: Unprocessed patient quotes are presented visually in the centre of the workspace.
- Drag-and-Drop Canvas: Users can drag quotes from the centre and drop them into distinct spatial sections on the page to form clusters, mimicking post-it notes.
- AI Collaboration UI: Interface elements that display AI-suggested theme names for user-generated clusters, alongside prompts to "Accept" or modify the AI's suggestions.
- Quote Traceability: Allows users to click on any clustered note to instantly read the exact, original survey response, eliminating the "black box" effect.

3.2 Backend (FastAPI)

Responsibilities:

- Manage the spatial coordinates and grouping metadata of quotes on the frontend canvas.
- Handle unstructured survey data ingestion and routing.
- Trigger the NLP pipeline once a user has clustered a sufficient threshold of quotes.

3.3 AI Thematic Pipeline

The AI acts as an assistant that learns from the user's manual actions rather than acting autonomously from the start.

- Phase 1: Observation (HITL Training): The system tracks how the user groups the initial set of quotes.
- Phase 2: Theme Naming: Once a cluster reaches a critical mass of quotes, the LLM analyses the semantic similarities within the group to suggest highly relevant theme names.
- Phase 3: Auto-Clustering (Inference): By learning the patterns from the initial user-defined themes, the AI evaluates incoming future survey responses and automatically suggests assigning them to the correct existing clusters. The user can accept, reject, or modify these AI suggestions.

4. Data Design

4.1 Entities

- Quote: Specific, anonymised patient text (acts as the individual "post-it note").
- Cluster: A user-defined spatial grouping of Quotes.
- Theme: The AI-suggested or user-defined semantic label applied to a Cluster.
- Metadata: Quantitative data linked to responses (e.g., age, region) used to overlay trends on the visual clusters.

5. Functional Requirements

The system should:

- Display unassigned quotes in a central visual queue.
- Allow users to drag-and-drop quotes into free-form clusters.
- Trigger AI theme name suggestions when a cluster reaches a predefined size.
- Automatically evaluate new quotes and route them to existing user-approved clusters.
- Maintain a transparent audit trail explicitly showing the rules or data weightings used to form themes.

6. Development Timeline

- Week 1: Foundation: Backend setup, state management for spatial grouping, and central quote ingestion queue.
- Week 2: Core Frontend: Implement the React-based drag-and-drop "post-it note" interface and Quote Traceability logic.
- Week 3: AI Integration: Connect the NLP pipeline to read user clusters, generate Theme Name suggestions, and implement the Auto-Clustering logic for new data streams.

7. Security and Data Privacy

7.1 UK GDPR & NHS Compliance

The system will keep sensitive patient feedback entirely within the organisation's secure IT environment, ensuring strict compliance with UK GDPR and NHS data protection standards.

8. Evaluation & Success Metrics

- Workflow Efficiency: Measuring the reduction in time taken to process a batch of 500 survey responses using the drag-and-drop tool versus traditional spreadsheet analysis.
- Auto-Clustering Accuracy: The percentage of AI-assigned new quotes that the user "Accepts" without needing to manually re-route them to a different cluster.

9. Infrastructure & Resource Allocation

9.1 Budget Strategy (£200 Allocation)

To ensure high availability while operating strictly within a £200 budget, computationally expensive AI tasks are decoupled from the web hosting infrastructure.

9.2 Compute Distribution

- University of Southampton Iridis HPC: Utilised for zero-cost, heavy batch processing and running the complex pattern learning algorithms required for the auto-clustering pipeline.
- Microsoft Azure: The £200 budget is strictly allocated to running an always-on secure Azure Virtual Machine to host the interactive FastAPI backend and React frontend, alongside secure, UK-region Azure Blob Storage for the anonymised survey data.